higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

CONCRETE STRUCTURES
NQF LEVEL 4

12 MARCH 2019

This marking guideline consists of 6 pages.

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QUESTION 1

1.1 Bars should be stored by size and length✓ in some logical manner and all tags at the same end for easy identification.✓ When bundles are opened the bar with the tag should remain in the bundle.✓ When bars are removed from the bundles, it is a good idea to cross out the number showing the original total number of bars and write the new number of bars left on the stock register.✓ (4)

1.2
1.2.1 Neutralise
1.2.2 Acidity
1.2.3 Limestone
1.2.4 Clay
1.2.5 Alkaline
1.2.6 Passive

(6 × 1) (6)
[10]

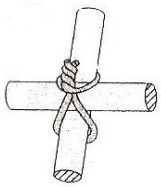
QUESTION 2

2.1 The embedded portion should not extend✓ into the specified minimum cover,✓which may affect the setting between the concrete and steel.✓ (3)

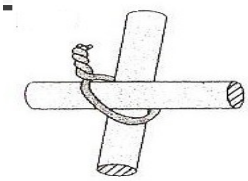
2.2
- Safety boots
- Safety clothes
- Safety goggles
- Gloves

(Any applicable 2 × 1) (2)

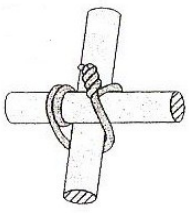
2.3 2.3.1



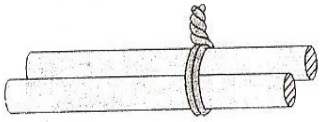
2.3.2



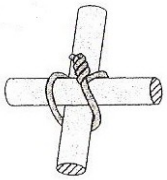
2.3.3



2.3.4



2.3.5



(5 × 2) (10)

2.4 2.4.1 False:✓ Black annealed✓ wire is used by most contractors because it is malleable and can easily be cut by hand (2)

2.4.2 False:✓ Ring slash ties are used to tie the vertical and horizontal bars together to prevent horizontal✓ bars slipping down (2)

2.4.3 True (1)

2.4.4 True (1)

2.4.5 False:✓ Epoxy-coated reinforcing bars must be fastened with epoxy or nylon-coated tie wire in order to protect the epoxy-coating from being damaged✓ (2)

2.5 2.5.1 It is a special pair of pliers used steel fixing to tighten wire✓ around reinforcement and to cut the wire ends. ✓ (2)

2.5.2 A bar tie twister is used to tighten bars✓ and make a twist in tie wire.✓ (2)

2.6
- The types of tie used and their position must be such that the required finish is achieved.
- Form ties and spacers left in position must not spoil the desired appearance or affect the durability of the structure.
- You must always neatly cut off and bend back the ends of the wires. This is to make sure that the embedded portion does not extend into the specified minimum cover –
- Ties should terminate at a distance of not less than the specified minimum cover from the formed surface of the concrete.
- Ties cast into spacer blocks should not extend deeper into the spacer block by more than half the depth of the spacer block. The concrete cover the projecting ends of ties or clamps must comply with specified concrete cover requirements.
- Reinforcement, including stirrups, links and tying wire, must not encroach into the specified cover. (Any applicable 4 × 2) (8)

[35]

QUESTION 3

3.1
- Dead load
- Impose load
- Environmental loads
- Hydrostatic loads

(Any 2 × 1) (2)

3.2 3.2.1
- 20: Number of bars √
- Y: type of bar √(high yield steel bar)√
- 10: size of bar /diameter of the bar√

(2)

3.2.2
- 4: Number of bars√
- R: type of a bar√ (Mild steel)√
- 8: size of the bar /diameter of the bar√

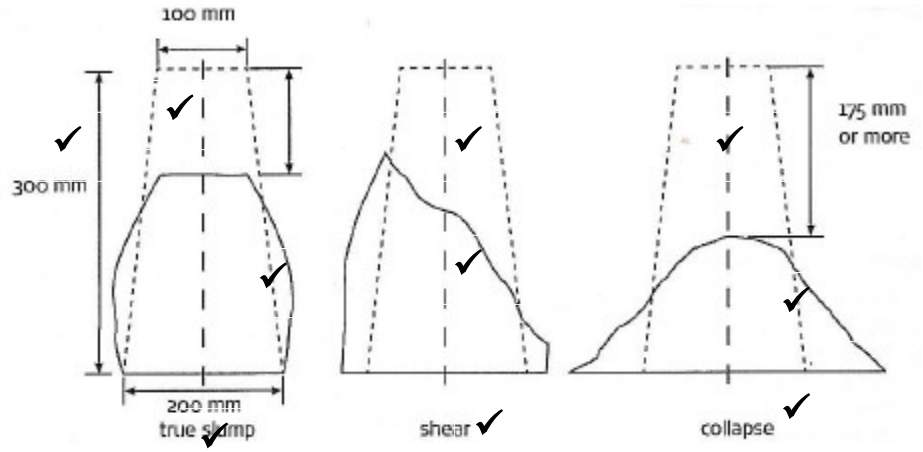
(2)

3.3 SANS 10100-2 states: "All reinforcement ,at the time of placing the concrete shall be free√ from rust,√ scale,√ oil√ and other coating√ that might reduce the bond between the steel√ and surrounding concrete√ or initiate corrosion of reinforcement."√ (8 x ½) (4)

[10]

QUESTION 4

4.1



(10 × 1) (10)

4.2 4.2.1 1,5 minutes (1)

4.2.2 Increase by 15 s for each additional cubic metre or part therefore (1)

4.2.3 10 minutes (1)

4.3 Reduce the temperature of the concrete mix✓
- Cool the mixing water by using well crushed ice for part or all of the mixing water. The ice particles have to be small enough to melt completely during the mixing process.✓
- Cool the aggregate by shading the stockpiles wetting the stone.✓
- Inject liquid nitrogen into the mix during mixing.✓

Add retarders to the concrete mix✓
- You can delay the stiffening of concrete by adding retarders to the mix.
- Retarders are chemicals that delay the setting of concrete✓ and work by physically inhibiting the hydration of cement. ✓
- Retardation is used to prevent cold joints in structures.✓

Add water reducers to the concrete mix✓
- Water reducers allow the concrete to become more plastics and with less water.✓
- The benefit is a strong concrete with less labour needed for placing and working it.✓
- High-range reducers or super plasticiser increases the concrete's ability to flow without reducing its strength.✓ (12)

[25]

QUESTION 5

5.1
- A company will not spend money on extra labour to repair damages
- A company will not lose productive time
- A company will not have to spend money on material for repairs
- There will be no risk of concrete being weakened by repairs
- There will be no unplanned hold-ups in the construction cycle
- There will be less risk of the concrete finish being substandard (6)

5.2 Surface plucking occurs when patches of the surface sticking to the formwork are pulled off the face during removal. (2)

5.3 Dry-pack mortar is a stiff sand-cement mortar that is typically used to repair small areas that are deeper than they are wide. (2)

[10]

QUESTION 6

6.1
- All formwork components need to be dry and clean before storing.
- Protect panels with waterproof covering.
- Do not store panels in a very hot and dry place.
- Do not store panels in direct sunlight. (Any applicable 3 x 1) (3)

6.2
- Passing traffic on site (construction vehicles or workers) could accidentally bump into and damage the forms.
- Construction workers could damage the forms by bumping or storing heavy objects on them.
- The forms could be damaged by rain.
- Wooden forms could warp in the sun.
- The forms could fall and be damaged. (Any applicable 2 x 1) (2)

6.3
- Components are easy to identify (correctly labelled).
- Do not block walkways.
- Are easy to get hold of.
- Are securely placed.
- Stored in a container with a lid so you can stack them on top of one another. (5)

[10]

TOTAL: 100

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